ERPsim for SAP HANA

Net Promoter Score (NPS) Reference Guide

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What is Net Promoter Score (NPS)?

NPS represents the percentage of customers who are likely to promote a company's products or services to their friends or colleagues. It has been introduced as "the number you need to grow" (Reichheld, 2003), and its use is common in the business due to its ease of implementation (Colvin, 2020). This measure uses referral behavior as a proxy to brand loyalty (Reichheld, 2011) and provides useful feedback for the quality and price of a company's products or services.

NPS is often measured by collecting responses to a question in the form of "How likely is it that you would recommend [company X] to a friend or colleague?" (Reichheld 2003, para. 23). Answers to this question are collected using a Likert scale from 0 (zero) to 10. In practice, NPS scores are collected from people in the company's target set of customers with no information on whether they use the products or services offered by the company or not.

In the interpretation of NPS, the respondents are categorized into three groups based on their responses (see Reichheld, 2011 for a more detailed discussion):

1. Promoters

   People who respond with a rating of 9 or 10 as their likeliness to recommend are considered in the 'promoters' group. These are people who have a positive attitude towards the company's services or products. Companies should try expanding this set of customers while retaining the existing customers in this group.

2. Passives

   People who respond with a rating of 6, 7, or 8 as their likeliness to recommend are considered in the 'passives' group. These customers have a neutral attitude towards the focal company. They are more likely to base their decisions on the price and quality of the product or service. Companies should try to improve the quality of their services so that the passives are turned into promoters.

3. Detractors

   People who give a rating of 5 or below are considered in the 'detractors' group. They have a negative attitude towards the company and are likely to use other brands or switch given a chance. Companies should find ways to make this set of people smaller by solving the issues that make some customers unhappy.

NPS Data in ERPsim

To achieve the pedagogical objectives of understanding and learning to analyse and interpret NPS data, we assume that the participants of the NPS survey are also asked whether they have bought the products supplied by the company. Thus, the NPS responses in the simulation can be divided into two groups: (1) responses from the users of the company's products or services (buyers), (2) responses from the users of other brands' products or services in the same category (non-buyers). The interpretations of these two data clusters might be different, so it is crucial to analyze the responses after separating these two groups (buyer vs. non-buyer).

NPS data in ERPsim can be retrieved into your analytics software of choice using OData. For more information on how to connect to OData, please refer to OData Services.

The NPS data can be accessed from the "NPS_Surveys" entity in the OData feed. In ERPsim, retail stores' consumers fill in NPS surveys, and the OData feed reports these responses' aggregate results in terms of their counts. Each line in the OData feed shows the count of responses from each retail stores for each product on each day for either the 'buyers' or the 'non-buyers.' These counts are presented in columns named SCORE_0, SCORE_1, …, SCORE_10 such that column SCORE_N refers to the count of respondents who gave N out of 10 in the Likert scale and having the same values for the attributes in Table 1. In other words, the SCORE_N values are counts of individual survey responses grouped by the attributes listed in Table 1.

Table 1. Unique Identifier Fields of OData feed "NPS_Surveys"

Field Name	**Description**
SIM_DATE	Simulation date as 'Round No'/'Day'
TYPE	Type of respondents: Buyer or Non-buyer
MATERIAL_NUMBER / MATERIAL_DESCRIPTION	Product code or name
CUSTOMER_NUMBER	Unique identifier of the retail store that collected the responses from their customers

The data feed includes several other fields, which may give additional information on the values of the attributes listed in Table 1. During the analysis, the data may be aggregated further using the columns that are not listed in Table 1. For example, since the decisions in the game focus on regions, e.g., north, south, and west, one may consider dropping the field named "CUSTOMER_NUMBER" by aggregating the counts over "AREA" using a summation function.

Calculation of the NPS Scores

NPS scores in ERPsim should be interpreted at four levels: product, simulation date, respondent location (area: west, east, or north), and respondent type (buyer vs. non-buyer). For each combination of values at these four levels, NPS can be calculated using this formula:

$$NPS_{ijkl} = \frac{|Promoters_{ijkl}| - |Detractors_{ijkl}|}{|Promoters_{ijkl}| + |Passives_{ijkl}| + |Detractors_{ijkl}|} \qquad (1)$$

where:

NPS_{ijkl} = NPS score for product i on simulation date j according to the responses from area k from respondent type l

$Promoters_{ijkl}$ = the set of responses that fall within the 'promoters' category for product i on simulation date j according to the responses from area k from respondent type l

$Passives_{ijkl}$ = the set of responses that fall within the 'passives' category for product i on simulation date j according to the responses from area k from respondent type l

$Detractors_{ijkl}$ = the set of responses that fall within the 'detractors' category for product i on simulation date j according to the responses from area k from respondent type l

i ∈ products; j ∈ dates; $k \in \{west, south, north\}$; $l \in \{buyer, non-buyer\}$; $|A|$: Size of set A

Example Calculation

Consider the output illustration in Table 2 for the OData feed named "NPS_Surveys." This illustration includes only the crucial columns for a simple interpretation of the NPS data in ERPsim. Other columns in the output may also be used for further analyses.

Let's assume the decision-maker wants to analyze the NPS responses concerning product 'milk' on simulation date '01/02' from 'non-buyers' located in the 'north' area. In this case, the decision-maker wants to calculate NPS_{ijkl} such that i =milk, j ="01/02", k =north, and l =non-buyer. There are two different rows in the example OData output for these attributes: rows 7 and 8 in Table 2. Each of these rows presents the responses collected by a different retail store in the same area. The decision-maker should sum the score values from these two rows to get an overall view of data from the whole north area.

Row	S_0	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8	S_9	S_10
7	0	0	0	0	1	0	0	0	1	4	5
8	0	0	0	2	0	0	1	0	1	0	0
Sum	0	0	0	2	0	0	1	0	2	4	5

After grouping the response sets from different retail stores, the number of promoters, passives, and detractors can be calculated. In this example with i =milk, j ="01/02", k =north, and l =non-buyer:

$$|Promoters_{ijkl}| = 4 + 5 = 9$$

$$|Passives_{ijkl}| = 1 + 2 = 3$$

$$|Detractors_{ijkl}| = 2$$

Therefore, using formula (1),

$$NPS_{ijkl} = \frac{9 - 2}{9 + 3 + 2} = 0.5$$

After calculating all possible NPS_{ijkl} values, a decision-maker may compare them to see the customers' brand loyalty concerning a product in a region. Using this information, the decision-maker may identify the respective strategies for several product-area pairs. An example illustration of the summary NPS table can be seen in Figure 1.

Table 2. An example of the data output of the OData feed "NPS_Surveys." Note: Columns "S_N" in this table illustrate columns named "SCORE_N" in the OData feed.

Row	SIM_DATE	TYPE	MATERIAL_ DESCRIPTION	CUSTOMER_ NUMBER	AREA	S_0	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8	S_9	S_10
1	01/02	Buyer	Milk	80305	West	0	0	0	0	0	0	0	1	2	2	0
2	01/02	Buyer	Milk	80296	North	0	0	0	0	0	0	0	0	1	1	1
3	01/02	Buyer	Milk	80300	South	0	0	0	0	0	0	0	0	0	2	2
4	01/02	Non-buyer	Milk	80299	South	0	0	0	0	0	1	1	0	0	0	0
5	01/02	Non-buyer	Milk	80304	West	0	0	0	0	0	1	1	0	0	0	0
6	01/02	Non-buyer	Milk	80305	West	0	0	0	0	1	1	1	0	0	0	0
7	01/02	Non-buyer	Milk	80296	North	0	0	0	0	1	0	0	0	1	4	5
8	01/02	Non-buyer	Milk	80294	North	0	0	0	2	0	0	1	0	1	0	0
9	01/03	Buyer	Cheese	80304	West	0	0	0	0	0	0	0	0	0	1	0
10	01/03	Non-buyer	Cream	80294	North	0	0	0	0	0	0	1	0	0	0	0
11	01/03	Non-buyer	Yoghurt	80304	West	0	0	0	0	0	1	1	2	1	0	0
12	01/04	Non-buyer	Ice Cream	80297	North	0	0	0	0	1	1	1	0	0	0	0

Net Promoter Score (NPS)

		MATERIAL_DESCRIPTION					
AREA	TYPE	Butter	Cheese	Cream	Ice Cream	Milk	Yoghurt
North	Buyer	0.857	1.000	0.000	-0.125	0.314	-0.423
	Non-buyer	-0.462	0.000	-0.333	-0.727	-0.906	-0.988
South	Buyer	1.000	1.000	0.500	-0.111	0.563	0.385
	Non-buyer	0.051	1.000	-0.067	-0.916	-0.946	-0.667
West	Buyer	0.500	0.667		-0.333	0.333	0.317
	Non-buyer	-0.545	0.563	0.033	-0.800	-0.942	-0.800

Figure 1. An example summary table including all possible NPS_{ijkl} values for comparison across products, regions, and customer types.

Interpreting NPS Scores in ERPsim

What Do NPS Scores Tell?

1. Buyers

The higher scores given by the buyers indicate high satisfaction and brand loyalty among the buyers of that product. Therefore, a high NPS score by the buyers may be a sign of customer satisfaction. On the other hand, a low score from a buyer may indicate that they purchased the products or services as a second option, and they may switch back once their first preference becomes available.

2. Non-buyers

The higher scores given by the non-buyers indicate that they are likely to purchase the products, given the chance. Their non-buyer status might have resulted from several factors, such as a too high price, a stockout of the product in the respective locations, or a better marketing policy implementation by another company in the same area.

How to Determine the Strategies According to the NPS Scores?

Case 1. NPS scores from buyers

An NPS score of 9 or 10 by the buyers indicates high satisfaction, the product may still be attractive to customers if the price is higher. Therefore, higher price levels could be tested for these products.

Medium scores from buyers, such as 5 or 6, indicate a slight risk of losing these customers. Not changing anything or reducing the prices could be reasonable strategies of managing these cases.

Low NPS scores from buyers indicate a higher risk of losing these buyers to the competitors. In such a case, a price discount might be considered. Likewise, the marketing spending can be increased for that specific product-region pair.

Case 2. NPS scores from non-buyers

NPS scores of 5 or higher from non-buyers indicate that the products are satisfactory for the potential customers, but there is something pushing the customers to other products. An increased revenue may be possible if this reason is identified and solved. First, it should be ensured that the customers can find the products on the shelves whenever they want. This can be done by increasing the regional transfer quantities after a thorough analysis of the inventory reports, sales quantities, and regional transfer parameters.

If there are no stockout situations in the regional warehouses but there are still non-buyers with high NPS scores, there are two possible alternative actions to make the products more attractive: (1) the marketing expenses can be increased for that product-region pair, (2) the prices might be reduced to attract these potential customers.

References

Colvin, Geoff. 2020. 'The Simple Metric That's Taking over Big Business'. *Fortune*. Retrieved 14 December 2020 (https://fortune.com/longform/net-promoter-score-fortune-500-customer-satisfaction-metric/).

Reichheld, Fred. 2011. *The Ultimate Question 2.0 (Revised and Expanded Edition): How Net Promoter Companies Thrive in a Customer-Driven World*. Harvard Business Review Press.

Reichheld, Frederick. 2003. 'The One Number You Need to Grow'. *Harvard Business Review*.